Smart Programming : YouTube Channel

An investment in Knowledge pays the best interest….

Multithreading Introduction in Java

=> Multitasking :-

➔ Performing multiple task at single time

➔ Examples :

- When we perform multiple task for example opening vlc, word, notepad, browser on single system etc

➔ Multitasking use the concept of context switching internally

context swiching meas the one process few second are execute and another process aslo few time execute one by excuting but same time two process are executing

➔ Multitasking can be achieved by 2 ways :-

1. Process based multitasking (Multiprocessing)
2. Thread based multitasking (Multithreading)

➔ Multitasking is used to reduce the ideal time of CPU.

the Mutli-tasking time the central processing unit are not ideal they are continuosly working case.

=> Multiprocessing :-

- ➔ Multiprocessing is the part when one system is connected with multiple processor (CPU)
- ➔ Multiprocessing is best suitable at system level or OS level

Mutli-processing means the one system are conected to the mutliple processor(CPU)

=> Multithreading :-

- ➔ Multithreading is the part which which multiple threads (sub-process) are executed at single time
- ➔ Examples :

1. Word
2. VLC
3. Games
4. Animations
5. Browser
6. Web-servers

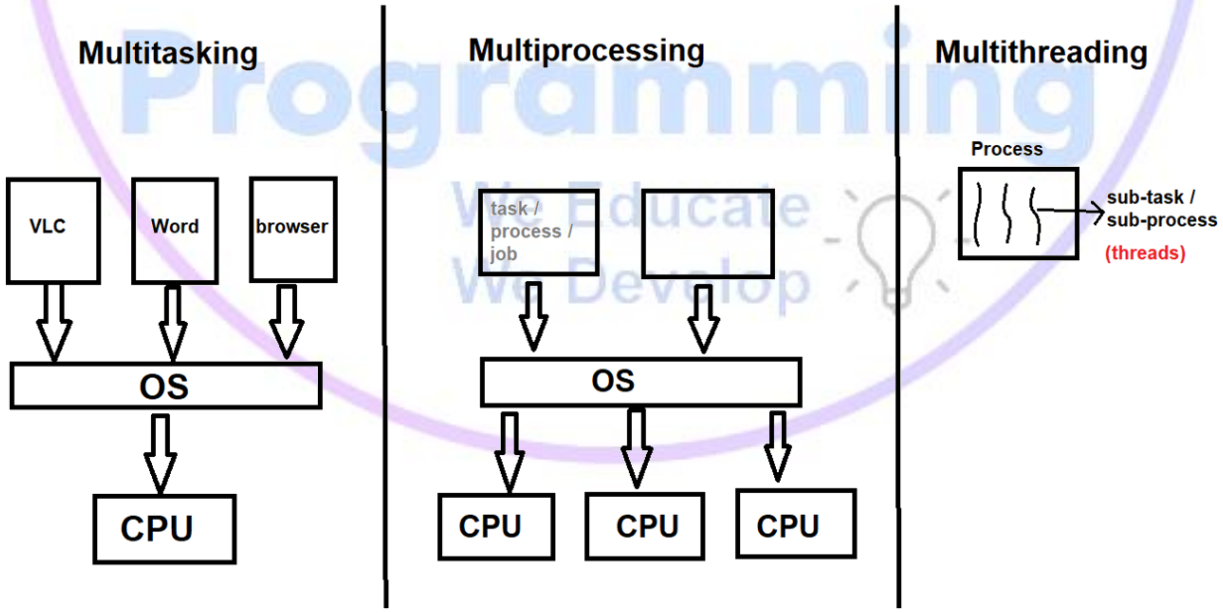
etc

➔ Multithreading is mostly implemented by java itself, we have to implement less part of multithreading.

➔ Java provides API for multithreading :-

1. Thread
2. Runnable
3. ThreadGroup
4. Concureency
5. ThreadPool
1. etc

API -> Application Program Interface



=> What is difference between Process & Thread ?

1. Process is the heavy-weight task performing multiple tasks
   Thread is the light-weight sub-process performing single task

2. Process does not depends on each other
   Threads depends on each other

3. Context switching is difficult in process (takes more time)
   Context switching is easy in threads (takes less time)

3. Process shares the different address space
   Threads shares the same address space

4. Process does not require synchronization
   Threads requires synchronization

=> How to create threads :

-> There are 2 ways to create threads :-

1. By using "Thread" class
2. By using "Runnable" interface

=> What is "Thread" :-

➔ Thread is the pre-defined class which is present in java.lang package

➔ Syntax :

```
class Thread implements Runnable
{
    //constructors

    //methods
    run()
    start()
    etc
}
```

=> What is "Runnable" :-

➔ Runnable is pre-defined interface present in java.lang package

➔ Syntax :

```
interface Runnable
{
    //method
    run()
}
```

=> Which is better way to create thread, Thread or Runnable ?

-> Runnable is better way to create threads in java because if we inherit Thread class then we were not able to inherit any other class as multiple inheritance is not supported in java but if we inherit Runnable interface then we can inherit more interfaces and can inherit the class also

Company Links & Contacts

Company Name: Smart Programming (+91 62838-30308)

Address : Chandigarh & Mohali (Punjab), India

Websites: https://www.smartprogramming.in/

https://courses.smartprogramming.in

Android App:
https://play.google.com/store/apps/details?id=com.smartprogramming

YouTube Channel:
https://www.youtube.com/c/SmartProgramming